

Context Compression Techniques for Legal Document Analysis

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Abstract

Large language models (LLMs) are increasingly applied to legal case analysis, where a single matter may span hundreds of pages across contracts, correspondence, expert reports, and court filings. The standard approach — retrieval-augmented generation (RAG) — addresses context size constraints by retrieving semantically similar document chunks, but fails on queries that require synthesizing facts distributed across multiple documents. This thesis proposes and evaluates *structured extraction* as a complementary approach: as case documents are ingested, key entities — parties, events, claims, damages, and deadlines — are extracted into a persistent, structured *FactSheet* that supplements chunk-based retrieval for case-level queries.

A LangGraph-based conversational agent implementing this architecture is evaluated against a RAG-only baseline across two legal datasets, multiple session stages, and three model configurations. Results show statistically significant improvements in correctness ($p = 8.5 \times 10^{-3}$) and relevancy ($p = 3.4 \times 10^{-4}$) without a proportional increase in token consumption or inference latency. A consistent and statistically robust reduction in score variance is also observed across all five evaluation metrics, suggesting that structured state reduces the brittleness introduced by retrieval failures. The performance gap is most pronounced for smaller, less capable models, indicating a practical cost-quality substitution: a cheaper model paired with the structured architecture can approximate the performance of a larger model on RAG alone. These findings demonstrate that schema-driven extraction is an effective and cost-efficient strategy for improving factual accuracy and reliability in multi-document legal settings.

Chapter 1

Introduction

The legal industry is increasingly adopting large language models (LLMs) to assist lawyers in navigating the complexity of modern litigation. Legal cases are inherently multi-document affairs: a single matter may involve contracts, correspondence, expert reports, court filings, and evidence spanning hundreds of pages. As a case develops, new attachments arrive, earlier positions are revised, and the relevance of existing documents shifts. The central challenge for an LLM-based legal assistant is not merely reading these documents — it is maintaining a coherent, accurate understanding of the case as a whole, across all its documents and all its turns of development.

The standard approach to this problem is retrieval-augmented generation (RAG): rather than loading all documents into context, the system retrieves the chunks most semantically similar to the user’s query. This is a practical solution to the context size problem, but it introduces a different and more fundamental limitation. Legal questions frequently require synthesizing facts that are distributed across multiple documents — a party’s identity established in one contract, a disputed event described in correspondence, a damages figure appearing in an expert report. Chunk-based retrieval cannot produce this synthesis: it operates on isolated local fragments, and high semantic similarity to a query does not guarantee that the retrieved chunks contain the relational context needed to answer it correctly. The consequence is that naive RAG agents perform well on direct factual lookups but poorly on case-level reasoning.

An alternative — injecting all documents directly into a large context window — does not solve the problem either. Even as model context windows have grown to one million tokens and beyond, empirical evidence shows that effective utilization degrades well before the nominal limit, and model performance deteriorates for information buried deep in a long context. Beyond capacity, loading an entire case file for every query is impractical and costly. The goal should instead be to provide the minimum amount of information that maximizes relevance to the query at hand.

This thesis proposes and evaluates a third approach: *structured extraction*. As case documents are ingested, key facts — parties, claims, events, damages, governing law — are extracted and organized into a structured *FactSheet*: a persistent, machine-readable summary of the case. The FactSheet supplements chunk-based RAG for queries that require a case-level overview, providing relational context that retrieval alone cannot assemble. It is initialized at project creation and updated incrementally as new documents arrive, so the agent always has access to a coherent picture of the case without re-reading the full document history.

This thesis addresses the following research questions:

Main RQ: *Does schema-driven cross-document extraction into a structured case representation improve the factual accuracy of an agentic RAG system in multi-document legal settings, without incurring proportional increases in query-time cost?*

RQ1: *To what extent does structured extraction mitigate the cross-document retrieval limitations of naive RAG in legal case analysis?*

RQ2: *Does the structured case representation maintain or improve agent performance as document volume grows across case sessions?*

RQ3: *Can a structured extraction architecture compensate for reduced model capacity — enabling a smaller, less expensive model to achieve performance comparable to a larger baseline?*

The system is implemented as a LangGraph-based conversational agent. The main RQ is evaluated through a controlled comparison of a structured-state agent against a RAG-only baseline across factual quality and resource consumption metrics. RQ1 is addressed by examining performance on queries that require cross-document synthesis. RQ2 is examined through session-level trajectories as document volume grows. RQ3 is addressed by comparing performance gaps across models of varying capability.

The contributions of this thesis are: (1) the design and implementation of initialization and incremental update pipelines for structured legal case state management, (2) an evaluation framework comparing structured state agents against RAG-only and baseline approaches across multiple models and datasets, and (3) an empirical evaluation of how structured extraction affects performance across datasets, model configurations, and document volumes.

The remaining chapters are organized as follows: Chapter 2 presents the theoretical background on LLM context limitations, RAG and its limitations, and structured extraction approaches. Chapter 3 describes the methodology, system architecture, and evaluation design. Chapter 4 presents experimental results. Chapter 5 discusses findings and implications, and Chapter 6 concludes with a summary and directions for future work.

Chapter 2

Theory

This chapter builds the theoretical case for the approach taken in this thesis. The argument unfolds in three movements: first, why large language models alone are insufficient for multi-document legal case analysis; second, why retrieval-augmented generation — the standard remedy — carries fundamental limitations that make it inadequate for cross-document reasoning; and third, what the literature suggests as more capable alternatives, which together motivate the structured extraction approach evaluated in this thesis.

2.1 Large Language Models and Context Windows

Modern LLMs are built on the transformer architecture [VSP⁺23], which processes input as a sequence of tokens bounded by a fixed context window. Recent models advertise increasingly large context windows — some exceeding one million tokens — which may appear to solve the problem of large document corpora. This appearance is misleading for at least three reasons.

First, the nominal context size overstates effective capacity. Hsieh et al. introduce RULER, a benchmark measuring the *real* usable context of long-context LLMs, and demonstrate that performance degrades substantially before the advertised limit is reached [HSK⁺24]. Second, LLMs do not attend uniformly across the context: performance is highest when relevant information is at the beginning or end, and degrades for information buried in the middle — the *lost in the middle* phenomenon [LLH⁺23]. Third, and most practically, a legal case involving dozens of attachments spanning hundreds of pages cannot be loaded in full for every query without prohibitive cost and latency, regardless of model capability.

The implication is a design principle: rather than maximizing the amount of text in context, the goal should be to include the *minimum* amount of information while ensuring that information is as relevant as possible to the query at hand. This principle rules out naive full-document injection and motivates retrieval as the primary mechanism for selecting what enters context. The question then becomes: what are the limits of retrieval?

2.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) operationalizes the minimum-relevant-information principle by retrieving only the most semantically similar document chunks at query time, rather than loading entire documents [LPP⁺21]. Documents are chunked and embedded offline into a vector store; at inference time, the top- k most similar chunks are retrieved and inserted into the LLM context alongside the query.

For straightforward factual lookup — “what is the defendant’s registered address?” — this works well. The answer lives in a single location in the document corpus, it is semantically similar to the query, and a chunk-based retriever will find it. The problems emerge when queries require understanding that spans multiple documents, or when the relevant information is not directly expressible as a similarity to the query string itself.

2.3 Limitations of Naive RAG for Complex Reasoning

2.3.1 Semantic Similarity Does Not Guarantee Relevance

The retrieval step selects chunks based on embedding similarity to the query. High similarity does not guarantee true relevance: documents may share keywords or surface features without addressing the underlying intent or logic of the query [GXZ⁺25]. In legal contexts, where questions about causation, liability, or damages require understanding relational chains across documents, surface-level matching is an insufficient retrieval criterion.

Compounding this, injecting all retrieved chunks directly into context can produce fragmented or contradictory input that confuses the LLM and degrades answer quality [GXZ⁺25]. The problem is not just what is retrieved but how it combines in context.

2.3.2 Cross-Document and Multi-Hop Reasoning

The more fundamental limitation is that the answer to a legal query often does not reside in one or two isolated chunks, but is distributed across multiple documents. Existing naive RAG systems are inadequate for queries that require retrieval and reasoning over multiple pieces of supporting evidence across documents [TY24]. In multi-document settings, relevant information is spread across sources and requires coherent synthesis: naively concatenating retrieved passages leads to fragmented or contradictory responses, a problem that is especially acute in legal and domain-specific QA where multi-hop reasoning is critical [LSL⁺25].

This limitation is structural, not a matter of tuning. A chunk embedding represents a local fragment of a single document. No similarity threshold or retrieval hyperparameter can make a chunk-based retriever produce a coherent synthesis of facts distributed across an entire case file.

2.3.3 Unstructured Retrieval and Missing Information

Standard RAG processes unstructured text passages without leveraging structured domain knowledge, and struggles with complex queries requiring multi-hop reasoning or domain-specific knowledge organization [JOJ⁺25]. Unstructured retrieval also risks returning passages that contain non-atomic information — facts entangled with irrelevant surrounding text — that can mislead the LLM rather than inform it [JOJ⁺25]. Chunk size amplifies this: small chunks provide insufficient context for understanding a passage, while large chunks dilute the semantic signal and retrieve off-topic content.

These three limitations — relevance mismatch, cross-document reasoning failure, and unstructured knowledge gaps — together define the core challenge this thesis addresses. A legal agent that relies solely on chunk-based RAG will, by design, struggle with case-level questions that require a coherent picture of the whole. This motivates approaches that move beyond isolated chunk retrieval toward structured, relational representations of case knowledge.

2.4 Graph-Based and Multi-Hop Retrieval

One line of work addresses RAG’s limitations by enriching retrieval with relational structure. GraphRAG [ETC⁺25] builds a knowledge graph over the document corpus, enabling community-level summarization and structured traversal of entity relationships rather than relying on chunk similarity alone. HopRAG [LWC⁺25] extends this with explicit multi-hop reasoning, constructing logical chains across documents to answer queries that require synthesizing evidence from multiple sources.

These approaches directly target the cross-document reasoning failure described above. Rather than asking whether a chunk is similar to a query, they ask how entities and facts across the corpus are related. The factsheet approach implemented in this thesis can be understood as a lightweight, domain-specific variant of this idea: rather than constructing a general-purpose knowledge graph, it extracts and organizes precisely the information most relevant to legal case management — parties, claims, events, damages, and governing law — into a structured, queryable representation that supplements RAG for case-level queries.

2.5 Agentic Context Management

The final theoretical component concerns how structured knowledge is maintained over the lifecycle of a multi-turn conversation. Rolling summarization — periodically condensing conversation history into a compact summary — is a practical strategy for preventing context overflow in long sessions [Mic24]. Stateful agent frameworks such as LangGraph enable persistent state management through checkpointing, allowing an agent to maintain structured representations that accumulate case knowledge across turns without requiring the full document history to remain in context [Lan24].

In the system developed in this thesis, the factsheet is the primary persistent state artifact. It is initialized during document ingestion and updated incrementally as new attachments are processed and new facts confirmed, providing the agent with a structured, token-efficient overview of the case at every point in the conversation. The theoretical argument, then, is that this persistent structure addresses what RAG cannot: a coherent, relational picture of the case that is available without vector-based retrieval: the factsheet is loaded directly from the structured database, not assembled from semantically matched chunks at query time.

The following chapter describes how these theoretical motivations are translated into system design and how their combined effect is evaluated empirically.

Chapter 3

Method

3.1 System Architecture

The system is organized as a `uv` workspace containing one core module and three auxiliary modules. The **agent** package is the primary contribution of this thesis: it implements the full backend, including the LangGraph agent, document processing pipelines, vector store integration, and database management. The remaining packages are supporting tools: **ui** provides a Streamlit interface for non-technical users to interact with the agent; **evals** contains the evaluation harness used to benchmark agent configurations; and **data-viewer** is a lightweight helper for inspecting and editing evaluation datasets. The architecture description that follows concerns the **agent** module exclusively.

3.1.1 Overview

The backend is implemented as a **FastAPI** application (`agent/main.py`) that exposes a RESTful API with Server-Sent Events (SSE) for real-time response streaming. The frontend is a **Streamlit** application that communicates with the backend over HTTP, consuming streamed responses. Authentication is handled via **Supabase Auth** using JWT tokens, which are validated on each request through a **HTTPBearer** dependency in FastAPI.

3.1.2 Agent Layer

The core agent is implemented as a LangGraph **StateGraph** (see Figure 3.1). The graph has three nodes:

- **init** — Injects the system prompt as the first **SystemMessage** on a new conversation; is a no-op on resumption.
- **call_llm** — Assembles the full input payload (system prompt, factsheet context, conversation history, and any per-request attachment content), sanitizes it for provider compatibility, then streams the LLM response.
- **call_tool** — Executes all tool calls emitted by the LLM in the previous turn, with per-call caching to avoid redundant invocations within the same session.

Control flows as: **START** → **init** → **call_llm**. A conditional edge then routes to **call_tool** if the LLM response contains tool calls (looping back to **call_llm** afterwards), or to **END** if not. To manage long conversations within the model’s context window, a rolling summarization strategy is applied every **sum_rate** messages (configurable, default 20) using a dedicated **Summarizer** module.

Conversation state is persisted using LangGraph’s **AsyncPostgresSaver** checkpointing, backed by **Supabase PostgreSQL**, enabling session resumption and multi-turn dialogue across requests.

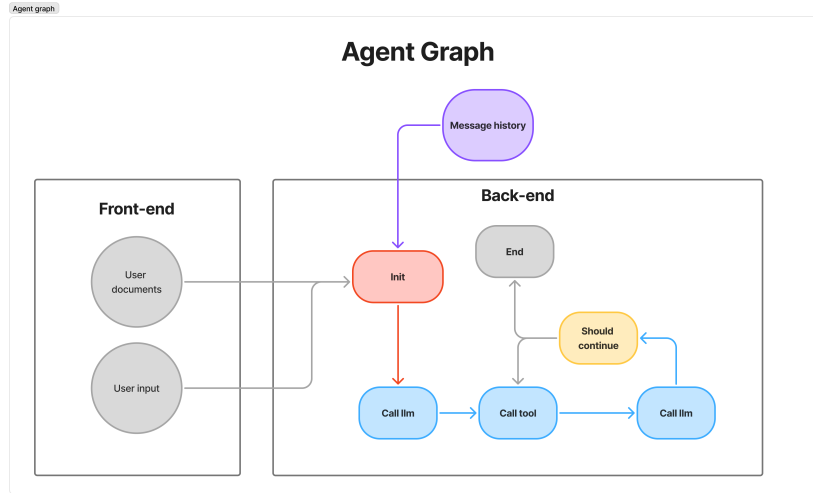


Figure 3.1: Agent graph: `init` → `call_llm` with a conditional loop through `call_tool`.

3.1.3 Tool Layer

The agent has access to a set of callable tools defined in `tools.py`. The full tool set (`TOOLS`) used in the primary agent configuration includes:

- **tavily_search** — Web search via the Tavily API.
- **read_attachments** — Fetches and returns the full parsed content of one or more documents by file ID. The parsed body is cached in the PostgreSQL database (populated during the document analysis pipeline); if not yet cached, the raw file is read from GCS and parsed on demand.
- **list_attachments** — Lists a project’s attachments and emails (with metadata such as title, date, and category), optionally filtered by date range and significance. Returns file IDs that the model can pass to **read_attachments**.
- **show_elements** — Returns structured factsheet elements (events, parties, claims, damages, deadlines) for a project, optionally filtered by date range and significance level.
- **query_project_attachments** — Performs semantic retrieval (RAG) against project-specific document chunks indexed in the vector store.
- **query_laws** — Semantic retrieval against a corpus of Norwegian legislation, with optional filtering by law title.
- **read_specific_law** — Retrieves exact paragraph text from legislation by title and paragraph number via a parameterised BigQuery query.

An ablated tool configuration, **BASELINE_RAG_TOOLS**, is defined for evaluation purposes. It includes web search, law retrieval (**query_laws**, **read_specific_law**), and project document RAG (**query_project_attachments**), but excludes the structured factsheet tools and direct file-reading access.

3.1.4 Data and Storage Layer

Document persistence is handled through two complementary mechanisms. Raw files (PDF, DOCX, PPTX, EML) are stored in **Google Cloud Storage (GCS)**, where they can be retrieved on demand by the **read_attachments** tool when no parsed body cache is available. Parsed and chunked document content is embedded and stored in a persistent **vector store** for semantic retrieval. Structured project data — including the **FactSheet** and session metadata — is stored in a **PostgreSQL** database via the **SupabaseManager** CRUD interface.

3.1.5 Init and Update Pipeline

Project initialization (POST /init-project) and updates (POST /update-project) are handled by the `ProjectPipeline` class, which compiles a LangGraph `StateGraph` for each operation (see Figure 3.2). The graph exploits LangGraph's fan-out semantics to run independent work in parallel:

- `collapse_emails` and `initialize_input` start in parallel from `START`. `collapse_emails` merges EML thread duplicates; `initialize_input` runs an LLM call to extract an initial structured summary from the user's case description.
- `extract_emails` follows `collapse_emails` and decodes email attachments from the collapsed threads.
- `parsing` and `storage` both fan out from `extract_emails`. `parsing` processes all documents (PDF/OCR, DOCX, PPTX, EML) into plain text; `storage` uploads raw files to GCS in parallel.
- `embedding` follows `parsing` and indexes chunked document content into the persistent vector store.
- `analyze` is a fan-in node that waits for both `parsing` and `initialize_input` to complete. It then dispatches all documents and emails to the LLM in size-bounded batches (see Section 3.2) to extract structured entities: attachments, emails, events, claims, damages, and deadlines.
- `update_metadata` follows `analyze` and runs an LLM call to update the case title, background, start date, and primary parties based on all extracted data.
- `qc_analysis` also fans out from `analyze` and runs a quality-control pass over the extracted entities, logging any inconsistencies or discrepancies found.
- `save` follows `update_metadata` and persists all extracted entities to Supabase PostgreSQL.

The update pipeline is identical in structure but adds a `load_project_data` node at the start that loads the existing factsheet before analysis, enabling incremental updates. Status events are streamed back to the frontend via SSE at each node transition.

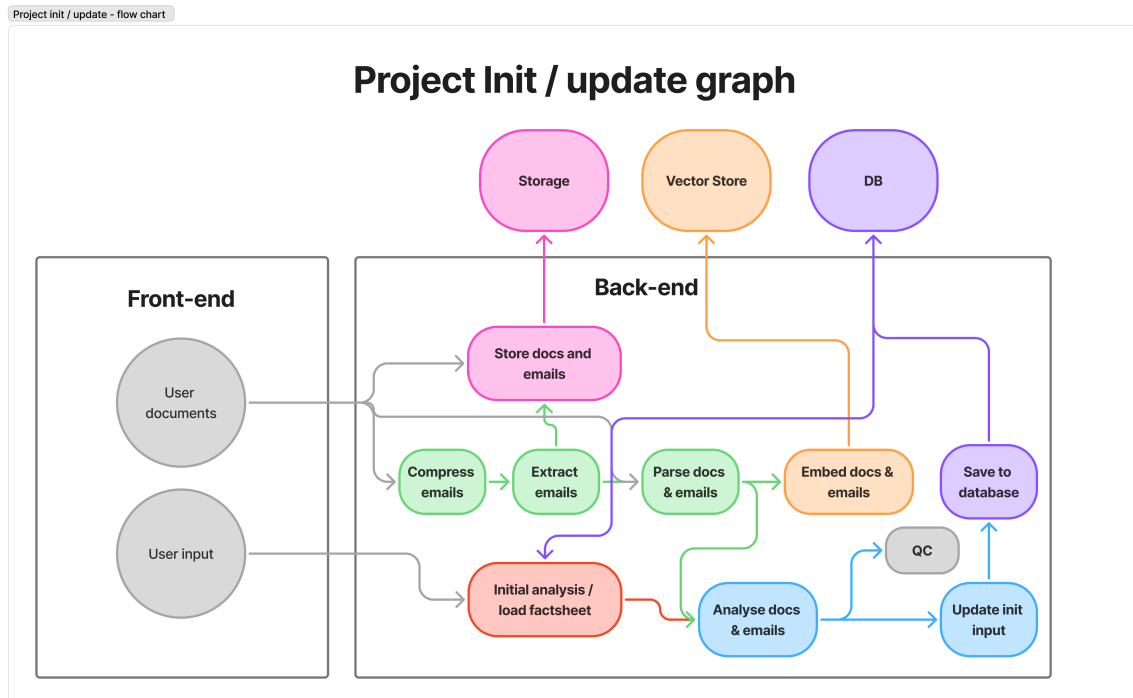


Figure 3.2: Init and update pipeline graph: parallel fan-out for parsing, storage, and initialisation, converging at the `analyze` fan-in node.

3.1.6 Clean Pipeline

The `ProjectClean` class provides a cleanup graph that can be triggered on demand (see Figure 3.3):

- **Clean Elements** (`compile_clean_elements`): `load_project` → `clean` → `save_results`. The `clean` node runs `_dedup_elements_node`, which dispatches all element types (events, parties, claims, damages, deadlines) to the LLM in configurable chunks concurrently via `asyncio.gather`. Crucially, the LLM is only asked to return the IDs of elements to keep — rather than rewriting full element objects — which minimises output token usage, reduces hallucination risk, and enables parallel deduplication of several hundred elements efficiently. Elements whose IDs are not returned are discarded, and results are saved back to the database.

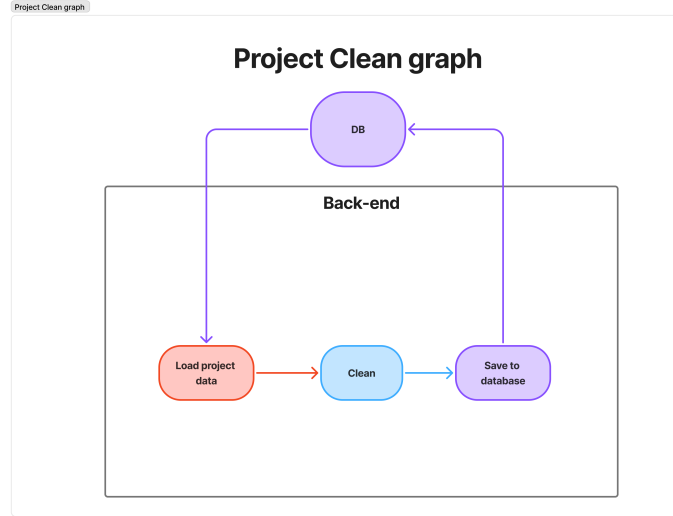


Figure 3.3: Clean pipeline graph: `load_project` → `clean` → `save_results`, used for both element deduplication and metadata rewriting.

3.1.7 LLM Configuration

The system supports multiple LLM providers through LangChain’s `init_chat_model` interface. **Google Gemini** serves as the primary provider, with **OpenAI GPT** available as a secondary option. The system also supports a range of open-source models from providers such as Qwen (Alibaba) and Zhipu AI, accessed through compatible APIs. The active model is configurable per session request, allowing for flexible experimentation and evaluation across providers and model families.

3.1.8 Tech Stack Summary

3.2 Design Choices

The system architecture was guided by several pragmatic design choices, prioritizing development speed, flexibility, and cost-efficiency given the constraints of a master’s thesis. By separating the descriptive architecture from the rationale, the core motivations for the selected technologies are outlined below.

Agent Layer and Orchestration

The core agent is implemented using **LangGraph** [Lan24]. This framework was chosen primarily for its fast build time, which is essential given the limited time available for this thesis. It provides a clean **StateGraph** abstraction that is highly expandable for future extensions, such as adding dedicated reasoning nodes or document re-ranking stages. Furthermore, LangGraph is model-agnostic, providing built-in support for multiple LLM providers. This flexibility is crucial, allowing the system to easily switch between models or add new ones as desired for evaluation and experimentation.

Layer	Technology
Language	Python 3.13
Agent framework	LangGraph (StateGraph)
LLM orchestration	LangChain (<code>init_chat_model</code>)
LLM providers	Google Gemini (primary), OpenAI GPT (secondary)
Backend API	FastAPI + Uvicorn
Streaming	Server-Sent Events (SSE)
Frontend	Streamlit
Authentication	Supabase Auth (JWT)
Database	Supabase (PostgreSQL)
Vector store	BigQuery Vector Store (persistent, cross-session)
File storage	Google Cloud Storage (GCS)
Conversation checkpointer	LangGraph <code>AsyncPostgresSaver</code>
Document parsing	PyPDF2, ocrmypdf, python-docx, python-pptx
Embeddings	Google <code>gemini-embedding-001</code>
Web search	Tavily

Table 3.1: System tech stack.

Database and Storage Architecture

The system’s persistence layer separates structured relational data, unstructured document storage, and semantic vector retrieval to optimize for specific access patterns and constraints.

Relational Storage (PostgreSQL via Supabase): PostgreSQL was selected for its flexibility and highly efficient reads across tables. The structured nature of the factsheet state requires strict enforcement of relationships through primary and foreign keys to maintain data integrity and orderly matching on IDs [Sup20]. An initial attempt was made using a NoSQL database (Firestore); while easy and fast to set up, the lack of an enforced schema led to a loss of overview, with state inconsistencies and errors accumulating over time. Additionally, performing complex reads and combining data across different entities proved expensive and inefficient in NoSQL.

Supabase was chosen as the PostgreSQL provider because it is affordable, open-source, and seamlessly integrates other necessary products like Authentication. Having these features included out-of-the-box makes the development process straightforward and fast.

Database Schema

The database schema is designed to support the agent’s state management and document metadata, while being flexible enough to accommodate future extensions. There are three main groupings of tables:

Project related tables
Projects
Parties
Events
Deadlines
Attachments
Emails
Claims (legal specific)
Damages (legal specific)

Session related tables
Sessions
Session Events
Session Attachments

Tables for storing document metadata and relationships.

Agent checkpoints
Checkpoints
Writes
Blobs

Tables for the agent’s memory and conversation history.

These tables are combined to form the factsheet state, which is the agent’s internal representation of the case. The factsheet is updated and referenced throughout the agent’s operation, and it is crucial that it is stored in a structured and consistent manner.

File Storage (Google Cloud Storage): Raw documents (PDF, DOCX, EML, etc.) are stored in Google Cloud Storage (GCS). There were no specific architectural requirements for this other than cost-efficiency and reliability. GCS is affordable, integrates cleanly with other Google Cloud services already used in the stack, and provides durable object storage without requiring any dedicated infrastructure.

Relation to Graph-Based Retrieval: The combination of a relational schema and a vector store gives the system a hybrid retrieval architecture that shares structural similarities with graph-based RAG approaches such as GraphRAG [ETC⁺25]. The relational tables — parties, events, claims, damages, and deadlines — function as typed entity nodes, each linked back to the source document through foreign-key relationships and a `source_id` field. These links act as structured edges connecting extracted knowledge to its documentary origin, enabling the agent to reason not just about individual facts but about which documents substantiate them. Semantic search over the raw document chunks (via the vector store) serves the same role as embedding-based traversal in graph retrieval: it handles queries where the relevant content is not captured by the structured entities. The factsheet can therefore be understood as a domain-specific, relational graph over the case — not a general-purpose property graph, but a structured knowledge base with typed entities and cross-document connections, backed by semantic search for unstructured content.

Vector Store and Embeddings: Semantic retrieval relies on **Google’s embedding model** (`gemini-embedding-001`). This model was specifically chosen for its strong multilingual capabilities; smaller and cheaper open-source models often struggle with RAG matching when queries or documents are not strictly in English, which is a critical requirement for handling Norwegian legal texts.

For the vector database, a two-tier approach is used. **ChromaDB** is utilized as an in-memory store for fast, session-scoped retrieval. For persistent, project-scoped retrieval, **BigQuery Vector Search** is employed. BigQuery was chosen because it is affordable and operates on an on-demand pricing model. The primary drawback of BigQuery is its high latency, meaning a significant amount of time is spent indexing and saving to the vector store. However, this trade-off is worthwhile: the alternative would require provisioning an always-running vector database server, which introduces high continuous costs and additional operational setup.

FastAPI and Server-Sent Events

The backend is exposed as a standalone REST API using **FastAPI** [Ram18] to decouple it from the frontend. **Server-Sent Events (SSE)** are used for streaming responses because the agent pipeline is long-running and asynchronous. SSE provides a responsive user experience by pushing incremental status updates and partial results without the complexity of WebSockets, which are unnecessary for this unidirectional communication flow.

Concurrency, Reliability, and Tool Efficiency

Several engineering decisions were made to ensure the system operates efficiently under the latency and throughput constraints imposed by external APIs and large document volumes.

Async-first architecture. The entire backend stack is built on Python’s `asyncio` event loop. FastAPI natively supports async request handlers, and all LangGraph pipeline nodes are declared as `async` coroutines. This allows the server to handle multiple concurrent sessions without blocking on I/O-bound operations such as LLM calls, database queries, or storage reads.

Semaphore-based concurrency control. To prevent overloading external services, four independent `asyncio.Semaphore` instances are used — one per resource class: LLM, database, storage, and vector store. Each semaphore has a configurable maximum concurrency value drawn from `config.toml` (e.g. `[async.llm] max_concurrent_requests = 20`). Pipeline nodes acquire the relevant semaphore before dispatching work, ensuring that bursts of parallel tasks are automatically throttled without busy-waiting or sleep loops.

Batch-wise and chunk-wise parallel processing. During project initialization, documents are grouped into batches before being dispatched for LLM analysis. Batching is driven by two configurable thresholds: a maximum byte-size per batch (`project.threshold`) and a maximum file count per batch (`project.max_attachments / project.max_emails`). Each batch is wrapped in an async task that acquires the LLM semaphore, and all tasks are dispatched concurrently via `asyncio.gather()`. The same chunk-wise strategy is applied during factsheet cleaning and deduplication: element lists (events, claims, damages, etc.) are split into fixed-size chunks (`project.clean.chunk_size`), each chunk is cleaned by a separate concurrent LLM call, and results are merged after all tasks complete. This approach keeps individual prompts within token limits while maximising throughput.

Rate limiting and 429 handling. To comply with provider-imposed rate limits and avoid HTTP 429 (Too Many Requests) errors, each LLM instance is wrapped with LangChain’s `InMemoryRateLimiter`. The limiter implements a token-bucket algorithm: tokens are refilled at a configurable rate (`async.llm.requests_per_second`) and a burst capacity (`async.llm.max_burst_size`) allows short spikes. A single stateful limiter instance is shared across all coroutines that use the same model, so the rate limit is enforced globally rather than per-request.

Retry mechanism with exponential backoff. Transient API errors (e.g. network timeouts or momentary 5xx responses) are handled by a dedicated `apply_retry()` helper. It wraps any LangChain chain with `with_retry()`, configuring a stop condition (`retry_attempts` from config) and exponential backoff with random jitter (`wait_exponential_jitter=True`). The retry decorator is applied *after* structured-output binding so that validation failures are also covered. If all retry attempts are exhausted, the error is propagated and logged.

TOML-based configuration management. All tunable parameters — concurrency limits, rate limiter settings, retry counts, chunk sizes, model endpoints, and storage bucket names — are centralised in a single `config.toml` file. On startup, the file is parsed into a fully typed `AppConfig` object built with `Pydantic` and `pydantic-settings`. Nested configuration sections map directly to typed dataclasses (e.g. `AsyncConfig`, `ProjectConfig`, `VectorstoreConfig`), providing IDE auto-complete, validation at load time, and a single source of truth for the entire application. A `get_app_config()` singleton cached with `@lru_cache` ensures the config is parsed only once per process.

Tool interface design — string IDs over full objects. Agent tools are designed to accept minimal, flat inputs: all tools receive only string identifiers (e.g. `project_id: str, ids: list[str]`) rather than full domain objects. This keeps the JSON schema presented to the LLM small, reduces the token footprint of tool calls, and makes it straightforward for the model to construct valid invocations. Internal data fetching (loading full attachment records, querying the database) is performed inside the tool implementation, hidden from the model entirely.

Context compression via filtering and display shortening. A key design concern for the custom agent is managing the size of the structured context injected into each LLM call. As a case accumulates material, a naive approach of injecting the full factsheet verbatim would cause token consumption to grow unboundedly. Two complementary compression mechanisms are applied to mitigate this.

First, the `show_elements` and `list_attachments` tools accept optional `significance` and `date_range` parameters. These filters are implemented as PostgreSQL RPC functions on the Supabase backend and are applied server-side before any data is returned to the agent, ensuring that only elements above a configurable significance threshold and within the relevant time window are included in the context.

Second, the string representation of each structured element is deliberately abbreviated. Rather than serialising the full object with all fields, a compact display format omits less informative fields and truncates long free-text descriptions. Together, these two mechanisms reduce the effective model input by approximately 70–80% compared to an unfiltered, full-fidelity serialisation of the same factsheet. This compression is a significant contributor to the finding that the custom agent’s average token consumption remains statistically indistinguishable from that of the simpler baseline agent despite carrying substantially richer case context.

3.3 Evaluation Design

3.3.1 Datasets

The evaluation relies on two datasets developed specifically for this thesis in collaboration with legal experts from the law firm BAHR AS. Each dataset represents a real legal case processed through the Norwegian court system.

THRD-2021 is based on a publicly available district court case. The authentic court record — comprising the final submissions from both parties and the court’s verdict — is used directly as part of the document set. The remaining case materials (correspondence, internal notes, technical reports, agreements, and supporting evidence) were not publicly available and were therefore synthetically reconstructed to match the factual matrix and legal reasoning documented in those authentic records. This hybrid approach provides a realistic document topology while respecting confidentiality.

TOSL-2024 is a fully authentic case for which explicit permission was obtained to use the original documents. It is substantially larger and more complex than THRD-2021: it involves more parties, a longer timeline, more contested legal issues, and a higher total document volume. The original case file includes close to a thousand emails; these were excluded from the evaluation dataset due to the prohibitive per-run cost and execution time they would introduce across repeated evaluation trials. Even after this exclusion, TOSL-2024 remains considerably larger and more complex than THRD-2021. Because TOSL-2024 is a real case with real documents, it more closely represents the conditions under which the agent would operate in production, and its results should therefore be given greater interpretive weight than those from THRD-2021.

Both datasets are organized to simulate the chronological lifecycle of a legal proceeding. Each case is structured into several *sessions*, each containing a sequence of *queries*. This structure is designed to evaluate two critical capabilities of the agent:

The datasets are organized to simulate the chronological lifecycle of a legal proceeding. Each case is structured into several *sessions*, each containing a sequence of *queries*. This structure is designed to evaluate two critical capabilities of the agent:

1. **Contextual Awareness:** The ability to maintain a coherent understanding of the case as it evolves through different stages of litigation.
2. **Analytical Accuracy:** Each query is paired with a "gold standard" expected output, defined by legal experts. These references serve as the ground truth for measuring correctness, relevancy, and completeness.

By following a chronological order, the sessions require the agent to handle increasing amounts of information and maintain consistency over time, reflecting the professional requirements of a legal practitioner managing an ongoing case.

3.3.2 Evaluation Limitations and Caveats

Several factors limit the generalisability of the evaluation and should be considered when interpreting the results.

Absence of a plain LLM baseline. A natural comparison point would be an ordinary LLM queried without any specialised architecture — no document processing, no vector store, no RAG, simply the raw case files passed as context. This approach was explicitly attempted. In practice, the aggregated size of a realistic legal case file — spanning contracts, correspondence, technical reports, and legal submissions — immediately exhausted the rate limits of every model tested, well before a meaningful response could be produced. The sheer volume of document content triggers the providers’ token-per-minute limits even for a single query, making repeated evaluation across sessions entirely infeasible. Feeding entire case files into a plain LLM context is therefore not a viable strategy for any of the models evaluated, and the minimum meaningful comparison is a RAG-enabled architecture. This is also why the evaluation compares the custom agent against a Baseline + RAG configuration rather than an unaugmented LLM.

Dataset size and the scalability trade-off. The evaluation uses a constrained number of cases and queries, as running a large-scale benchmark across multiple model-agent pairs with repeated trials becomes prohibitively expensive and time-consuming. This is, however, in tension with a fundamental property of the system: the agent is designed to improve with larger and richer document sets, as more material increases the factsheet’s coverage and the quality of RAG retrieval. Evaluating on small datasets therefore likely *underestimates* the performance advantage of the custom agent in production-scale deployments.

Heterogeneity between cases. The two evaluation datasets differ substantially in origin, complexity, and document volume. THRD-2021 uses a hybrid document set (authentic court record supplemented by synthetically reconstructed case materials), while TOSL-2024 consists entirely of real documents. TOSL-2024 is considerably more complex: it covers more parties, a longer contested timeline, and a larger total document volume. This asymmetry means that performance differences across datasets may partly reflect document complexity rather than agent capability alone. However, since the agent is specifically designed for complex, large-scale cases, TOSL-2024 is the more ecologically valid test of its capabilities, and the dataset-level breakdowns in the results should be read with this in mind.

Exclusion of pipeline costs from resource metrics. The resource metrics reported in this evaluation — token usage and inference latency — cover only query-time activity: the LLM calls made by the agent when responding to a user query. The one-time cost of running the initialization and update pipeline (document parsing, entity extraction, embedding, and storage) is deliberately excluded.

This exclusion is methodologically correct for two reasons. First, including pipeline costs in a per-query resource comparison would produce a misleading figure on the evaluation datasets. Each dataset contains a small number of evaluation queries; amortising the full extraction cost across, say, four queries makes the cost per query appear orders of magnitude higher than it would be in practice. A production deployment handles hundreds or thousands of queries per case, over which the fixed extraction cost becomes negligible. There is no principled way to choose an amortisation denominator within the scope of this evaluation, and any choice would be arbitrary.

Second, the pipeline does not run at query time. It is an offline preprocessing step that can be triggered asynchronously — overnight, during off-hours, or on document arrival via integrations with document management systems and email. The practitioner does not wait for it. The relevant cost comparison is therefore query-time cost, which is what the reported metrics reflect.

Limitations of expected-answer evaluation. Model responses are graded against expected answers produced in collaboration with legal experts at BAHR AS. This approach carries two inherent risks: the expected answers themselves may contain inaccuracies, and — more fundamentally — many legal queries are open-ended, such that a correct and complete model response may differ substantially from the reference without being wrong. Three mitigations were applied to reduce this risk:

1. **Fact-based scope:** The evaluation is restricted to factual, document-grounded questions (e.g. "Who are the parties?", "What damages are claimed?"), deliberately excluding normative or predictive legal questions where correct answers are inherently ambiguous.
2. **Validatable content:** Questions are formulated so that their answers can be directly verified against the case documents, reducing the space for legitimate variation.
3. **Minimal-requirement framing:** Expected answers are written as *minimum* acceptable responses rather than exhaustive ideal answers, so a model response that covers the required facts is scored as correct even if it includes additional relevant information.

3.3.3 Evaluation Environment

Evaluation Framework:

The evaluation will be conducted using **DeepEval** [Con23], a framework for evaluating LLM outputs against reference answers. DeepEval allows for automatic scoring based.

Agent Configurations:

Two agent configurations are evaluated:

- **Custom Agent:** The full agent configuration with access to all tools, including structured factsheet access (`show_elements`, `list_attachments`), direct file reading (`read_attachments`), and RAG capabilities.
- **Baseline + RAG Agent:** An ablated version with access to web search, law retrieval, and project document RAG, but without direct file access or structured factsheet tools.

A plain LLM baseline without document retrieval was considered but excluded from the evaluation for the reasons described in Section 3.3.2: the volume of case material immediately saturates provider rate limits, making repeated evaluation impossible.

Model Configuration:

Since the scope of the thesis is not to evaluate the models themselves, but rather the design and architecture of the agent, a selection of four models is used for comparison. These include both closed-source and open-source variants to ensure the agent’s performance is consistent across different provider architectures.

Table 3.2: Overview of Large Language Models used in evaluation.

Model Name	Type	Total Params	Active Params	Provider
Gemini 2.5 Pro	Closed	Unknown	Unknown	Google
Gemini 2.5 Flash	Closed	Unknown	Unknown	Google
Qwen 3.5 [YLY ⁺ 25]	Open (MoE)	397B	17B	Alibaba
GLM-5 [GTZ ⁺ 26]	Open (MoE)	435.2B	40B	Zhipu AI

The evaluation will be conducted across all datasets using consistent model-agent pairings. To account for the inherent stochastic nature of LLM outputs, each evaluation set (model + agent configuration) will be executed 3 times to ensure statistical significance in the results.

3.3.4 Metrics

The evaluation will focus on several key metrics, measuring both the quality of the agent’s responses and its efficiency in terms of resource usage. Metrics that are to be evaluated are:

- **Correctness, Relevancy, and Completeness:** These semantic metrics are evaluated using the DeepEval framework. The agent’s output is compared against a "gold standard" reference created in collaboration with legal experts. Each metric is scored on a scale from 0 to 1.

- **Self-consistency:** The degree to which the agent’s output remains consistent across multiple queries and over time as the case evolves.

$$\text{Self-Consistency} = \frac{2}{n(n-1)} \sum_{i=1}^{n-1} \sum_{j=i+1}^n \cos(\mathbf{e}_i, \mathbf{e}_j) \quad (3.1)$$

where n is the number of generated outputs, $\mathbf{e}_i$ and $\mathbf{e}_j$ are the sentence embeddings of the i -th and j -th output, and the cosine similarity is defined as:

$$\cos(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|} \quad (3.2)$$

- **BERTScore:** A metric that evaluates the similarity between the agent’s output and a reference text using BERT embeddings [ZKW⁺20].
- **Passrate:** The percentage of queries for which the agent’s output meets a predefined quality threshold.
- **Efficiency Metrics:** **Token Usage** (total tokens consumed) and **Inference Time** (latency) are recorded to assess the practical viability of the system.

Correctness, relevancy, Completeness are all evaluated against the expected out (constructed in collaboration with legal experts) using DeepEval. The framework rates each response on a scale from 0 to 1, with a certain threshold for what is considered a ”pass” for the passrate metric.

BERTScore is calculated based on the expected output. Self-consistency is evaluated by comparing the agent’s responses across multiple queries and sessions to assess whether it maintains a coherent understanding of the case over time. Token usage and inference time are recorded during the agent’s operation.

3.3.5 Research Question Operationalization

Each research question is operationalized through a specific subset of the evaluation evidence collected.

Main RQ — accuracy improvement without proportional cost increase. The main RQ is evaluated by jointly examining quality metrics (correctness, relevancy, completeness, passrate) and resource metrics (input tokens, inference duration) across all queries and sessions. A positive finding requires both a statistically significant quality advantage for the custom agent *and* the absence of a significant resource cost increase. The one-sided Mann-Whitney U tests described below provide the formal statistical basis.

RQ1 — mitigation of cross-document retrieval limitations. RQ1 is addressed by examining the quality metrics specifically on queries that require synthesis across multiple documents and sessions. The dataset-level breakdown (Section 4) isolates TOSL-2024, which contains a larger document volume and more contested cross-document relationships, as the primary test case for cross-document reasoning. A larger performance gap on TOSL-2024 compared to THRD-2021 is interpreted as evidence that structured extraction specifically addresses the multi-document limitation.

RQ2 — performance stability across growing document volume. RQ2 is addressed by inspecting session-level performance trajectories. The datasets are organized chronologically into multiple sessions, each adding new documents. A custom agent that maintains or improves quality across sessions while the baseline degrades or stagnates provides evidence that the structured representation scales gracefully with document volume.

RQ3 — model capacity compensation. RQ3 is addressed by examining performance per model. The evaluation includes models of varying capability, from smaller open-source models (Qwen3 MoE 3B active parameters) to larger closed-source models (Gemini 2.5 Flash, Claude Sonnet). If the custom agent enables a smaller model to reach performance levels comparable to a larger baseline model, this constitutes evidence that structured context can partially substitute for raw model capacity.

3.3.6 A Priori Performance Expectations

The following expectations, grounded in the architectural differences between the two configurations, motivate the experimental design and provide a reference frame for interpreting the results.

Quality metrics over session progression. The central hypothesis is that the performance trajectories of the two architectures diverge as a case progresses. In the earliest sessions, the document set is small and context volume is modest; under these conditions the simpler RAG agent is expected to perform adequately, and the difference between architectures should be minimal. As subsequent sessions introduce new documents, parties, claims, and a growing timeline, the RAG approach becomes increasingly strained: retrieved chunks do not necessarily contain sufficient context for questions whose answers are distributed across multiple documents. The custom agent, by contrast, maintains a continuously updated factsheet and can draw on both structured case state and the same vector store as the baseline. The expected result is a positive or stable trajectory for the custom agent and a flat or degrading trajectory for the baseline.

Token usage over session progression. Token usage is expected to grow for both architectures as the case accumulates material, but for structurally different reasons. For the baseline, growth reflects increasing retrieval volume as more chunks are returned from a larger corpus. For the custom agent, growth reflects increasing *richness* rather than raw volume: the factsheet grows denser, but context compression keeps the effective input size in check. Output tokens are likewise expected to grow over time for the custom agent as richer context enables more thorough answers — a positive signal rather than an efficiency cost.

Inference time. Response latency is expected to increase with case progression for both architectures, proportionally to token growth.

3.3.7 One-sided Hypothesis Tests

Inspection of the metric distributions reveals pronounced left-skew with heavy concentration near 1.0 (see Figure 3.4), violating the normality assumption required by parametric tests such as the t -test. All hypothesis tests are therefore conducted using the one-sided **Mann-Whitney U test**, a non-parametric rank-based test that assesses whether one distribution tends to produce stochastically larger values than another, without assuming normality.

For quality/performance metrics (correctness, relevancy, completeness, passrate, self-consistency, BERTScore):

$$H_0 : \eta_{\text{custom}} \leq \eta_{\text{baseline+RAG}} \quad H_1 : \eta_{\text{custom}} > \eta_{\text{baseline+RAG}}$$

For cost/efficiency metrics (token usage, inference time):

$$H_0 : \eta_{\text{custom}} \geq \eta_{\text{baseline+RAG}} \quad H_1 : \eta_{\text{custom}} < \eta_{\text{baseline+RAG}}$$

Notation:

- η_{custom} = population median for the custom agent
- $\eta_{\text{baseline+RAG}}$ = population median for the baseline with RAG agent

All tests are one-tailed / one-sided.

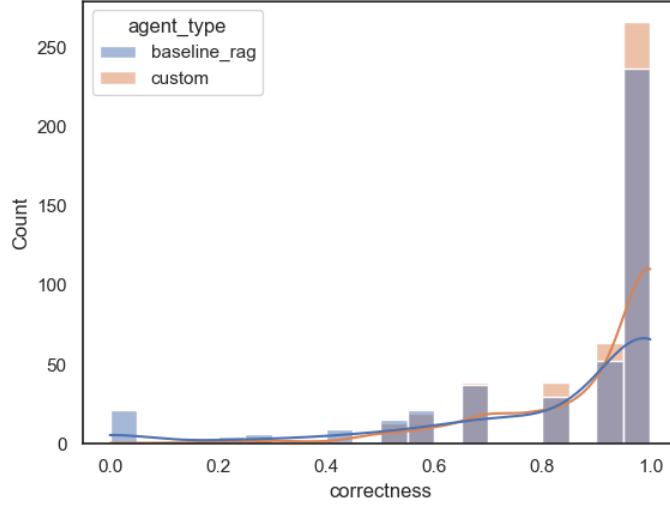


Figure 3.4: Distribution of correctness scores across all queries and sessions. The pronounced left-skew and heavy concentration near 1.0 violate the normality assumption, motivating the use of non-parametric Mann-Whitney U tests.

3.3.8 Levene’s Test for Equality of Variance

The Mann-Whitney U test evaluates whether one distribution tends to produce stochastically *larger* values than another, but says nothing about whether the two distributions differ in *spread*. To test whether the custom and baseline agents exhibit significantly different score variability, Levene’s test for equality of variance [Lev60] is applied to each quality metric.

The test statistic is:

$$W = \frac{N - k}{k - 1} \cdot \frac{\sum_{i=1}^k N_i (\bar{Z}_{i.} - \bar{Z}_{..})^2}{\sum_{i=1}^k \sum_{j=1}^{N_i} (Z_{ij} - \bar{Z}_{i.})^2} \quad (3.3)$$

where:

- k = number of groups (here $k = 2$: custom and baseline)
- N = total number of observations across both groups
- N_i = number of observations in group i
- $Z_{ij} = |Y_{ij} - \bar{Y}_{i.}|$, the absolute deviation of observation j in group i from its group mean
- $\bar{Z}_{i.}$ = mean of Z_{ij} within group i
- $\bar{Z}_{..}$ = overall mean of all Z_{ij}

Under the null hypothesis of equal variances, W follows an F -distribution with $k - 1$ and $N - k$ degrees of freedom. A significant result ($p < 0.05$) rejects the null and indicates that the two groups differ significantly in spread. The hypotheses are:

$$H_0 : \sigma_{\text{custom}}^2 = \sigma_{\text{baseline+RAG}}^2 \quad H_1 : \sigma_{\text{custom}}^2 \neq \sigma_{\text{baseline+RAG}}^2$$

Levene’s test is applied rather than Bartlett’s test because it is robust to departures from normality [Lev60] — a necessary property given the non-normal, left-skewed score distributions confirmed by Shapiro-Wilk ($p < 10^{-18}$ for every metric).

Chapter 4

Results

4.1 Quality Metrics

4.1.1 DeepEval: Correctness, Relevancy, and Completeness

The joint distribution of correctness, relevancy, and completeness across all queries and sessions is shown in Figure 4.1.

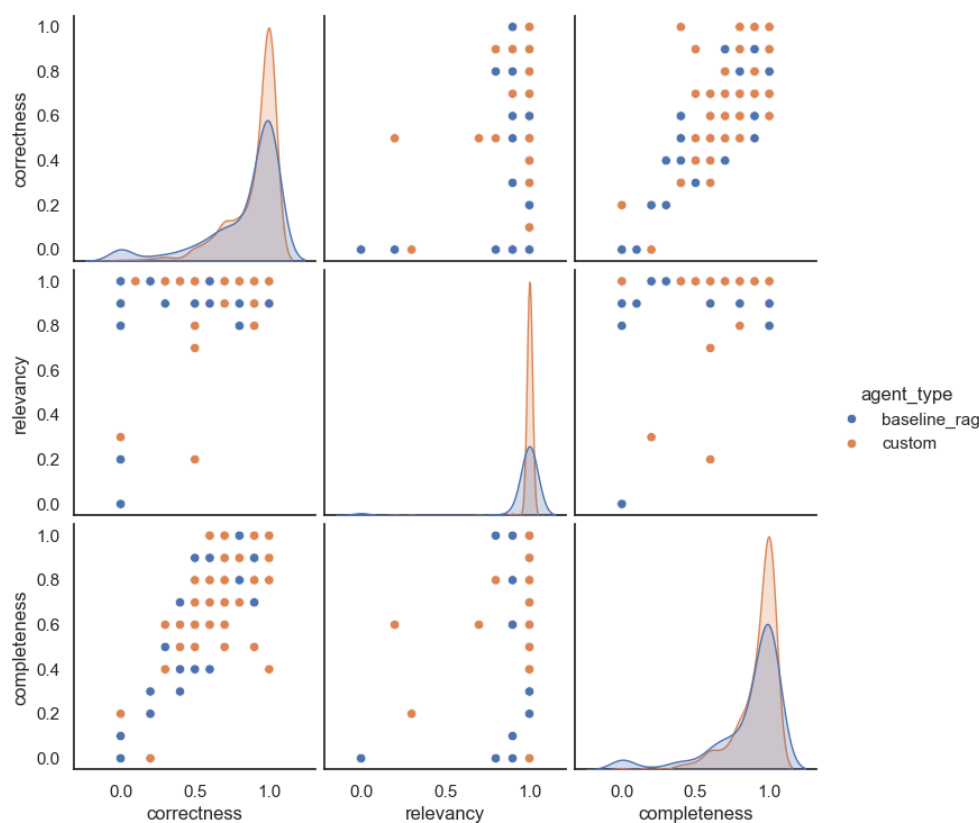


Figure 4.1: Pairplot of DeepEval quality metrics (correctness, relevancy, completeness). Diagonal panels show per-metric density estimates. The custom agent (orange) exhibits higher correctness and completeness, while relevancy is similarly high for both configurations.

Both distributions are left-skewed across all three metrics, with density concentrated near 1.0 — but the degree of concentration differs substantially. The custom agent’s distributions are more tightly clustered near the upper end, while the baseline’s are more spread, with a heavier low-scoring tail. This

confirms that the custom agent’s advantage is not driven by a small number of high-scoring queries but reflects a broad shift in the distribution. Relevancy scores are highly concentrated near 1.0 for both configurations — as long as an agent attempts to answer the question, the response tends to be on-topic regardless of architecture — though the baseline’s low-scoring tail corresponds primarily to cases where retrieval fails entirely. Completeness follows the same pattern: both are left-skewed, but the custom agent’s distribution is more concentrated near 1.0, while the baseline exhibits more low-completeness responses. Overall, the pairplot reveals greater consistency and reliability in the custom agent’s responses across all three quality dimensions.

Session-level trajectories tell the same story longitudinally (Figure 4.2).

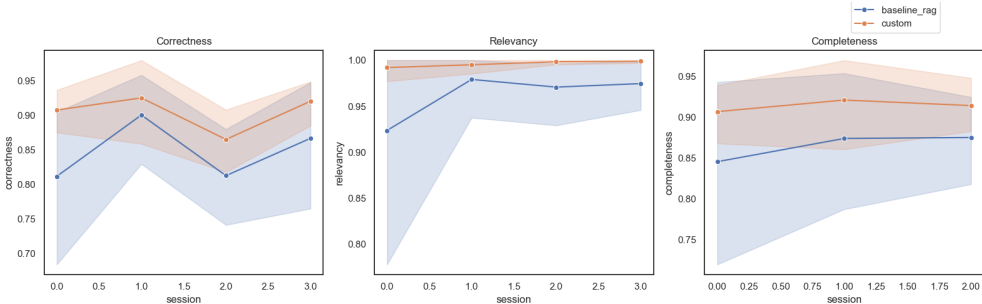


Figure 4.2: DeepEval quality metrics across sessions for each agent configuration.

Both agents broadly track each other through the session progression, but the custom agent remains consistently above the baseline throughout. Crucially, the variance of the custom agent’s trajectory is substantially lower: where the baseline exhibits wider swings between sessions, the custom agent maintains a more stable performance profile. This session-level consistency reinforces the pairplot finding that the quality advantage is broad and reliable rather than driven by occasional strong queries.

Breaking the correctness trajectory down by dataset (Figure 4.3) reveals that the performance gap is not uniform across cases.



Figure 4.3: Correctness over session progression, split by dataset.

On TOSL-2024 — the fully authentic case with the larger document volume and greater factual complexity — the custom agent’s advantage grows more noticeably as sessions accumulate. On THRD-2021, the custom agent also leads, but the gap is smaller and more stable. This pattern is consistent with the architectural hypothesis: the factsheet and structured retrieval tools confer the greatest benefit precisely when the case is large and complex, where naive RAG retrieval is most prone to missing relevant facts dispersed across many documents.

The model-level breakdown (Figure 4.4) adds a further dimension.

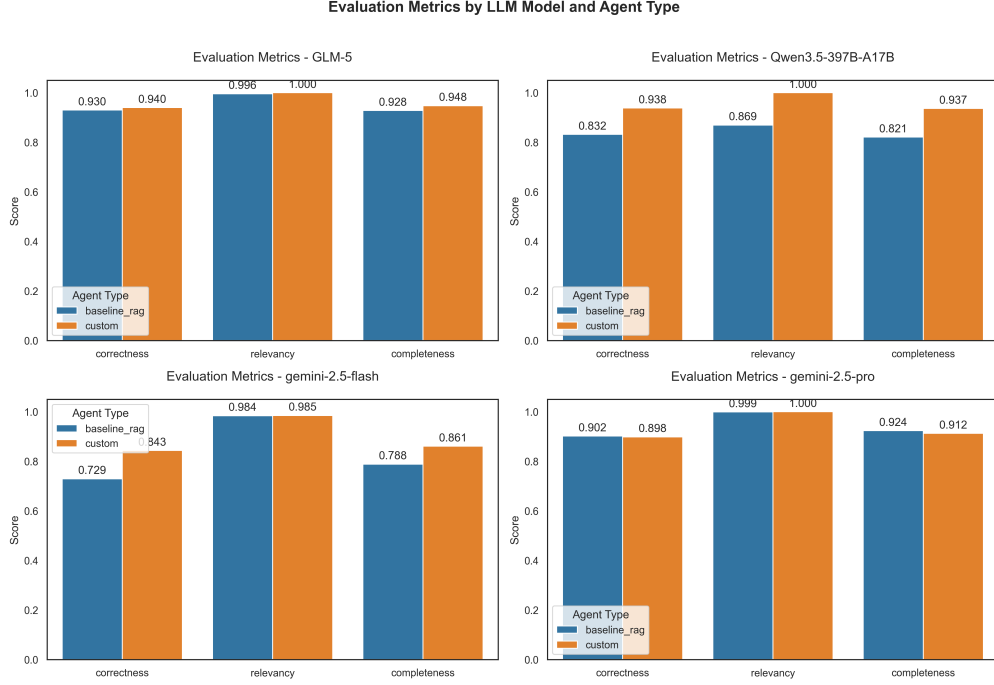


Figure 4.4: DeepEval quality metrics broken down by underlying LLM. The custom agent outperforms the baseline for every model tested, with the gap narrowing for the most capable model.

The custom agent outperforms the baseline across all three metrics for every model tested, with one exception: for the most capable model in the evaluation, the two configurations converge and the difference becomes negligible. For the smaller and less capable models, the margin is wider. This suggests that structured factsheet context acts as a capability amplifier whose marginal effect is greatest when the base model has limited capacity to compensate for noisy or incomplete retrieval. The implications of this model-dependent effect are discussed further in Chapter 5.

4.1.2 BERTScore

BERTScore measures surface-level semantic similarity to the reference answer and is therefore less sensitive to structured factual completeness than the DeepEval metrics. A response can score well on BERTScore while omitting specific facts that correctness and completeness evaluation would penalise.

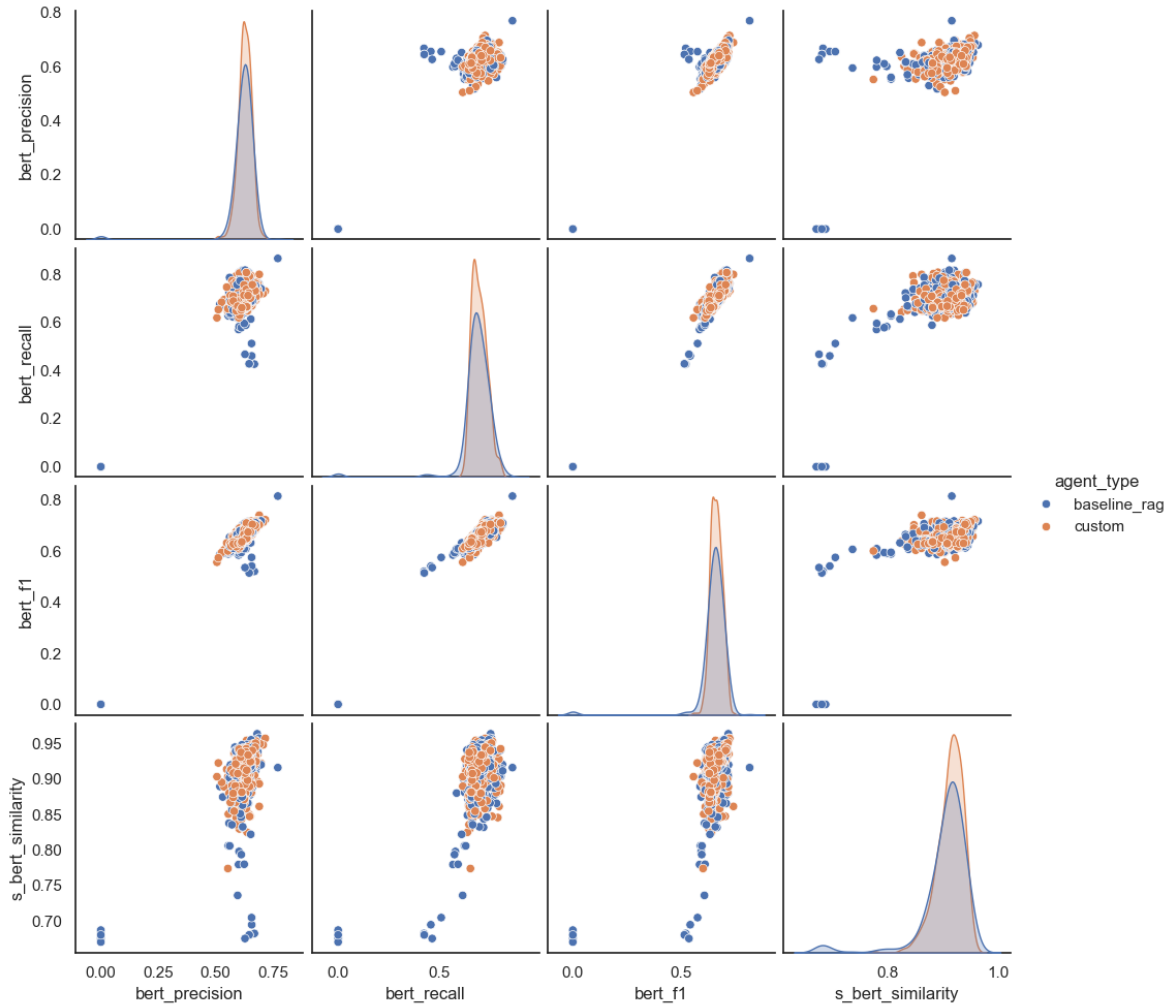


Figure 4.5: Pairplot of BERTScore sub-metrics (precision, recall, F1, and sentence-BERT similarity). Distributions for the two agent configurations largely overlap, indicating comparable surface-level semantic quality.

The distributions have considerable overlap between the two agent types, but the custom agent shows a higher peak with less variation and more consistent surface-level semantic similarity to the reference answers. The baseline has a wider spread with more low-scoring outliers, suggesting it occasionally produces responses less semantically aligned with the reference. The near-identical BERTScore distributions complement rather than contradict the DeepEval findings: both agents produce semantically coherent output, but the custom agent more reliably includes the specific factual content the gold-standard reference requires.

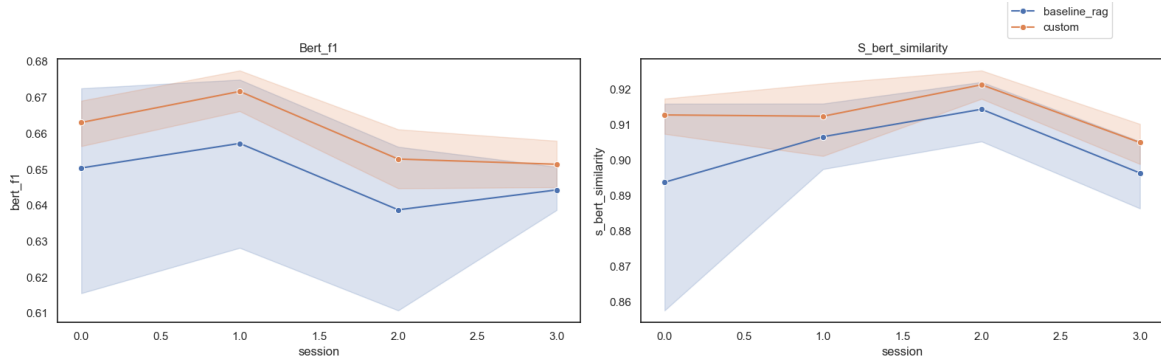


Figure 4.6: BERTScore F1 and sentence-BERT similarity across sessions. Trajectories are flat and closely aligned between configurations, with the custom agent showing marginally narrower confidence intervals.

Across sessions, BERTScore F1 and sentence-BERT similarity are relatively flat and closely aligned for both agent types, with only minor fluctuations. This is expected: surface similarity to reference answers does not change systematically as the case grows, since the phrasing of correct responses does not become more or less reference-like with document volume. The session-level divergence observed in the DeepEval metrics is therefore not an artefact of reference phrasing but reflects genuine differences in factual coverage.

4.1.3 Resource Metrics

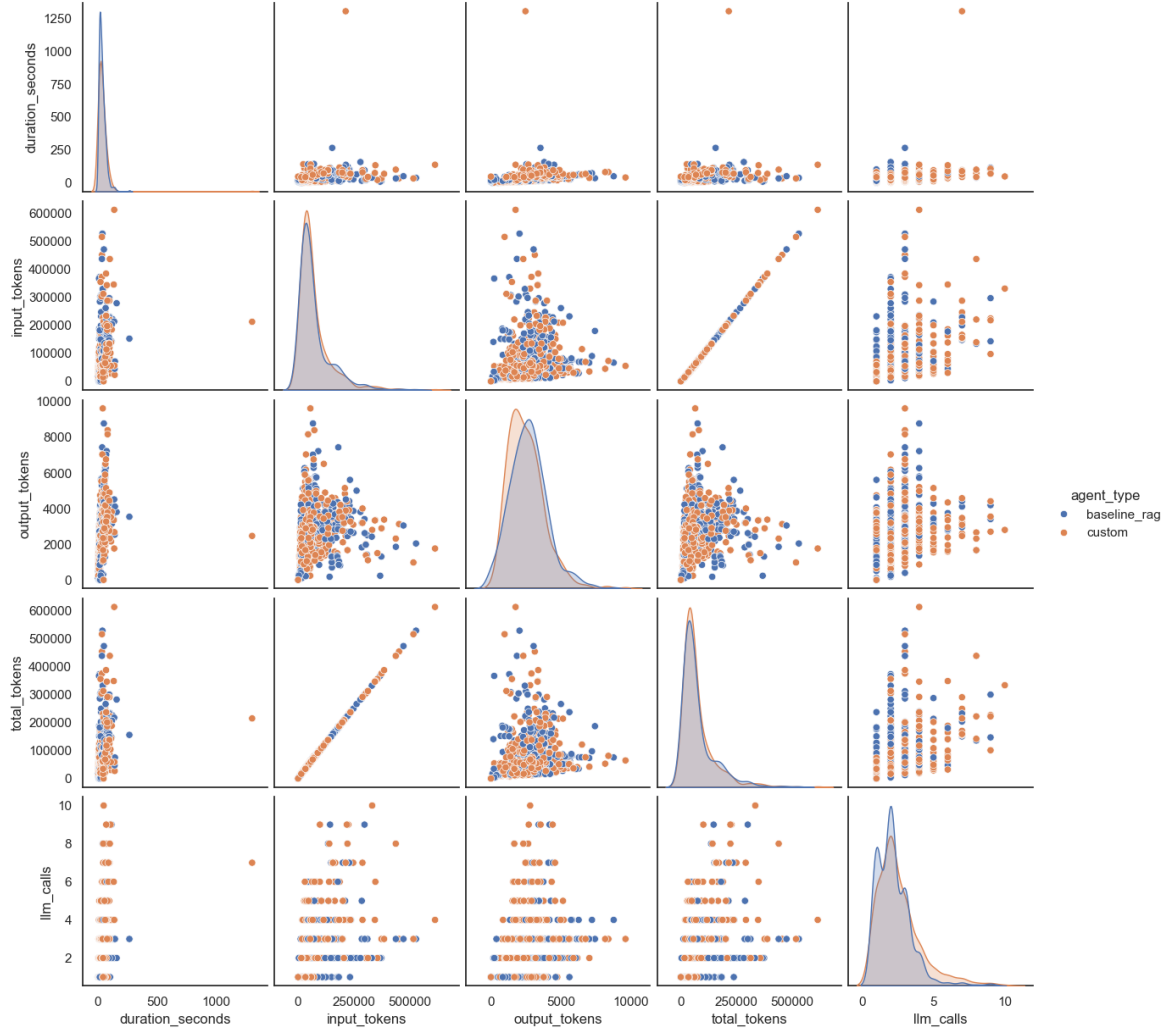


Figure 4.7: Pairplot of resource metrics (duration in seconds, input tokens, output tokens, total tokens, and LLM call count). The custom agent exhibits higher input token usage reflecting factsheet growth, while inference duration is more tightly bounded than the baseline.

The most notable pattern in Figure 4.7 is in input token usage. The custom agent spans a wider range, with some queries consuming upwards of 200,000 input tokens — consistent with the factsheet growing denser as the case accumulates material and vector retrieval returning richer chunks. The baseline clusters at lower input token counts, reflecting its simpler retrieval strategy, though it too exhibits high-token outliers for sessions with large document sets. Output tokens are similarly distributed, with the custom agent showing a slightly wider tail toward higher values, in line with the expectation that richer context enables more detailed answers. Inference duration is generally lower and more tightly concentrated for the custom agent; the baseline exhibits heavier tails at high latency, likely due to its unbounded document ingestion strategy. LLM call counts are comparable between configurations, typically in the range of 1–3 calls per query.

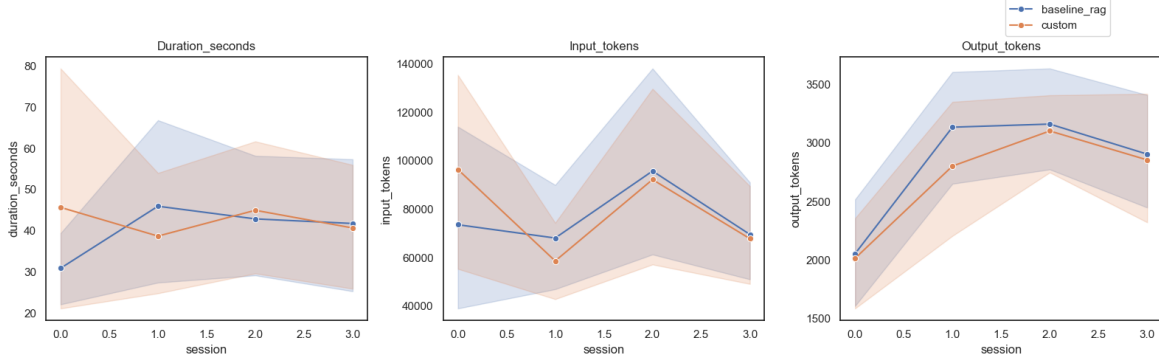


Figure 4.8: Resource metrics across sessions. Token counts and latencies fluctuate without a systematic trend for either configuration, and the two track each other closely throughout.

Unlike the quality metrics, no consistent trend is visible in the resource trajectories over session progression. Token counts and latencies fluctuate without a systematic increase, and the two configurations track each other closely throughout (Figure 4.8). The absence of a growth trend is partly explained by the context compression mechanisms applied to the custom agent: as the factsheet grows, significance filtering and display shortening keep the payload from expanding proportionally.

4.2 Statistical Modeling

4.2.1 Mann-Whitney U Test

All hypothesis tests use the one-sided Mann-Whitney U test, appropriate given the non-normal distributions of all metrics (Shapiro-Wilk, $p < 10^{-18}$ for every metric tested). The null hypothesis in each case is that the custom agent’s scores are not stochastically greater than those of the baseline agent.

Quality metrics. Table 4.1 summarises the results. Correctness and relevancy both show statistically significant advantages for the custom agent at the $\alpha = 0.05$ level. Completeness does not reach significance, though the directional trend is the same.

Table 4.1: Mann-Whitney U test results for quality metrics (one-sided, H_1 : custom $>$ baseline).

Metric	U-statistic	p-value	Result
Correctness	105,587.00	8.49×10^{-3}	Significant ($p < 0.05$)
Relevancy	101,782.50	3.42×10^{-4}	Significant ($p < 0.05$)
Completeness	38,820.00	8.34×10^{-2}	Not significant

The relevancy result warrants a cautious interpretation. Relevancy is almost universally high for both agents whenever a response is attempted: as long as the agent engages with the question, the answer tends to be on-topic regardless of architecture. The significant difference is primarily driven by cases where the baseline fails to retrieve any relevant documents and consequently produces no substantive answer, incurring a near-zero relevancy score. **Correctness is the most diagnostic quality metric:** it is hardest to satisfy and most sensitive to the architectural differences under evaluation, and the significant result ($p = 8.49 \times 10^{-3}$) carries the most interpretive weight.

The absence of a significant completeness result likely reflects the smaller number of completeness observations in the dataset and the inherent difficulty of distinguishing partial from complete answers when both agents frequently cover the required facts.

Resource metrics. Neither input token usage ($U = 99,750$, $p = 0.730$) nor inference duration ($U = 93,705$, $p = 0.163$) differs significantly between configurations. Mean input token counts are almost identical (custom: $72,286 \pm 77,114$; baseline: $71,968 \pm 75,645$) and mean durations are within one second of each other (custom: 35.1 s; baseline: 36.1 s). The custom agent achieves its quality improvements without imposing a measurable additional cost in tokens or latency.

4.2.2 Levene’s Test for Equality of Variance

While the Mann-Whitney U test establishes that the custom agent’s scores tend to be stochastically *higher*, it says nothing about the *spread* of those scores. Levene’s test for equality of variance addresses this directly: it tests whether the two configurations exhibit significantly different score variability, without assuming normality.

Table 4.2: Levene’s test results for equality of variance across quality metrics (custom vs. baseline+RAG). A significant result indicates the custom agent has lower variance.

Metric	Levene statistic	p-value	Result
Correctness	18.29	< 0.001	Significant
Relevancy	11.00	0.0009	Significant
Completeness	9.14	0.0026	Significant
BERTScore F1	8.00	0.0048	Significant
Sentence-BERT sim.	14.69	< 0.001	Significant

The test returns a significant result for every metric, including the complementary metrics (BERTScore and sentence-BERT similarity) where the mean difference between configurations is small. This means the custom agent is not only more accurate on average — it is *more consistent*. The baseline agent’s score distribution is significantly wider across all five metrics, with a heavier low-scoring tail that represents cases of partial or failed retrieval. The custom agent’s reduced variance implies more predictable, reliable responses: fewer failures at the low end, and less dependence on whether retrieval happens to return the right chunks for a given query.

In a legal setting, this reliability dimension may be as practically significant as the mean improvement. A system that occasionally fails unpredictably is harder to trust and supervise than one that performs steadily, even if its average score is similar.

4.2.3 Self-consistency



Figure 4.9: Distribution of self-consistency scores by agent type. The custom agent shows a higher median and narrower interquartile range.

Self-consistency scores reinforce the pattern seen in the DeepEval metrics. The custom agent exhibits a higher median and a narrower interquartile range than the baseline, indicating more consistent responses across repeated queries. The baseline shows a wider spread, with more low-scoring outliers, suggesting greater variability in how it handles similar questions across sessions.

4.3 Summary by Research Question

Table 4.3 summarises the empirical evidence for each research question.

Table 4.3: Research question outcomes based on empirical results.

RQ	Finding	Evidence
Main	Custom agent improves factual accuracy without significant cost increase	Correctness significant ($p=8.5\times 10^{-3}$); token usage and latency statistically equivalent
RQ1	Evidence is directionally consistent but inconclusive; a larger gap on the more complex dataset is suggestive but rests on few sessions and does not hold uniformly	Wider gap on TOSL-2024 sessions 3–4 only; THRD-2021 shows no clear difference
RQ2	Custom agent maintains stable or improving quality across sessions; baseline performance is more variable and lower throughout	Session trajectory plots; narrower confidence intervals and higher median for custom agent at later sessions
RQ3	Structured context amplifies smaller model performance; the gap between custom and baseline agents narrows as model capability increases	LLM barplot: gap is widest for smaller models, negligible for the most capable model

Main RQ. The results support an affirmative answer: the custom agent achieves statistically significant improvements in correctness and relevancy over the RAG-only baseline, while consuming a statistically equivalent number of tokens and producing responses within the same latency envelope. The quality gain is not purchased at a proportional resource cost.

RQ1. The evidence for RQ1 is directionally consistent with the hypothesis but not strong enough to support a firm conclusion. THRD-2021 shows no clear performance gap between configurations. TOSL-2024 shows a somewhat wider gap in sessions 3–4, but this rests on a small number of sessions and does not hold uniformly across all metrics. The data is too thin to isolate whether the effect is driven by cross-document synthesis demands specifically or by other case-level differences in complexity and document type. RQ1 remains open and would require a targeted benchmark of explicitly synthesis-demanding queries to resolve conclusively.

RQ2. Session-level trajectories show that the custom agent maintains quality across all sessions with narrower variance, while the baseline exhibits wider swings and a less stable profile. This supports the conclusion that the structured representation scales with document volume rather than degrading under it.

RQ3. The model-level breakdown provides the clearest evidence for RQ3: the custom agent’s structured context delivers the largest marginal gain for smaller and less capable models, with the gap shrinking to near-zero for the most advanced model tested. This suggests that the factsheet acts as a capability compensator — partially substituting for the reasoning capacity that larger models supply natively.

Chapter 5

Discussion

5.1 Research Question Findings

This section interprets the empirical results in relation to each research question, drawing on the statistical evidence from Chapter 4 and situating the findings within the theoretical context established in Chapter 2.

Main RQ: accuracy improvement without proportional cost. The results support an affirmative answer. The custom agent achieves statistically significant improvements in correctness ($p = 8.5 \times 10^{-3}$) and relevancy ($p = 3.4 \times 10^{-4}$) while consuming a statistically equivalent number of tokens and inference time compared to the RAG-only baseline. The context compression mechanisms — significance filtering and display shortening applied server-side before context injection — are largely responsible for keeping token usage in check as the factsheet grows: rather than injecting the full structured state verbatim, the agent receives a filtered, abbreviated representation tailored to the query context. The combined picture is that schema-driven extraction improves factual accuracy at no measurable additional cost per query.

Beyond the mean improvement, the variance reduction is arguably the most consistent finding in the evaluation. Levene’s test for equality of variance is significant across all five metrics tested — correctness, relevancy, completeness, BERTScore F1, and sentence-BERT similarity — including the complementary metrics where the mean difference is modest. The baseline agent’s score distributions are significantly wider, with a heavier low-scoring tail that reflects the brittleness of pure retrieval: when the relevant chunks do not rank highly for a given query, performance drops sharply. The custom agent’s structured state provides a reliable fallback that absorbs these retrieval failures, yielding more predictable responses across the board. In a professional legal setting, this consistency advantage may carry more practical weight than the mean improvement alone: a system that occasionally fails unpredictably is harder to trust and integrate into a practitioner’s workflow than one that performs steadily.

RQ1: mitigation of cross-document retrieval limitations. The evidence for RQ1 is suggestive but not conclusive, and the conclusion should be stated carefully. THRD-2021 shows no clear performance gap between the two architectures. TOSL-2024, the larger and more complex authentic case, shows a somewhat more pronounced advantage for the custom agent in the later sessions (sessions 3–4), but this pattern rests on a small number of sessions and does not hold uniformly across all metrics. With only two datasets and relatively few sessions each, the data is too thin to isolate whether the difference reflects document complexity, query-type composition, or genuine cross-document synthesis demands.

A further theoretical nuance is worth noting: naive RAG is not straightforwardly *volume-sensitive* in the way the original framing implies. Because retrieval always returns a fixed top- K regardless of corpus size, a larger document pool does not mechanically degrade retrieval quality. What RAG genuinely struggles with is *synthesis queries* — questions whose answers are distributed across multiple chunks that cannot all fit within a single top- K result. This is a query-type limitation, not a volume limitation. The evaluation did not explicitly control for the proportion of synthesis queries across

sessions and datasets, making it difficult to attribute the observed TOSL-2024 pattern specifically to cross-document retrieval failure rather than to other case-specific factors.

In summary, the results are directionally consistent with the hypothesis that structured extraction mitigates cross-document retrieval limitations, but the empirical basis is insufficient to support a strong conclusion. Confirming RQ1 would require a larger benchmark with sessions explicitly designed around synthesis-demanding queries.

RQ2: stability across growing document volume. Session-level trajectories show that the custom agent maintains consistent quality with narrow variance throughout case progression, while the baseline exhibits wider fluctuations and a less stable profile as sessions accumulate new documents. This is consistent with the design hypothesis: the factsheet absorbs new information incrementally at each session boundary, so the agent’s knowledge of the case becomes richer rather than more fragmented as volume grows. The baseline, by contrast, must retrieve from an ever-larger document pool with no persistent structure to guide retrieval, making its performance increasingly dependent on whether the relevant chunks happen to rank highly in a given semantic search.

RQ3: compensation for reduced model capacity. The model-level breakdown provides the most striking result for RQ3: the performance gap between the custom and baseline agents is largest for the smaller models and shrinks to near-zero for the most capable model tested. For smaller models, the structured factsheet provides the key facts in an organised, pre-digested form, reducing the reasoning burden on the model itself. A more capable model can compensate for noisy or incomplete retrieval through superior inference, making the factsheet less decisive. This model-dependent pattern has a direct practical implication for cost management: a smaller, less expensive model paired with the custom architecture can approximate the performance of a larger, more expensive model on the RAG-only baseline, suggesting a cost-quality substitution that may be valuable in production deployments.

5.2 Value Beyond Performance Metrics

The evaluation results demonstrate a performance advantage for the custom architecture as case volume grows, but the headline metrics alone understate the practical value of the design. Even in a hypothetical scenario where correctness and completeness scores were identical across configurations, the custom architecture would remain preferable in a production legal setting for reasons that no automatic metric captures.

The factsheet as a product artifact. The baseline agent’s output is ephemeral: it produces a text response and leaves no persistent structured representation of the case behind. The custom agent, by contrast, produces and continuously maintains a structured factsheet as a natural side-effect of normal operation. This factsheet—comprising parties, events, claims, damages, deadlines, and document metadata—is a machine-readable, queryable representation of the case that can be consumed directly by frontend systems without any additional LLM inference. Concrete applications include timeline visualisations, party graphs, claims dashboards, and deadline trackers, all of which can be rendered from the factsheet with no extra cost.

Human oversight and auditability. The structured state enables a qualitatively different relationship between the lawyer and the system. Rather than trusting a black-box text response, a practitioner can inspect the factsheet directly, verify individual entries against source documents, and correct errors at the element level. If the agent produces an incorrect answer, the source of the error can often be traced to a specific factsheet entry—a mis-extracted event date or an omitted party—and corrected there, improving all subsequent responses. This kind of targeted, human-in-the-loop correction is not possible with a stateless RAG agent, where errors are buried in the implicit context and must be addressed by re-uploading or re-prompting.

Versioned case understanding. Because the factsheet is updated incrementally at each session boundary, it constitutes a natural audit trail of how the agent’s understanding of the case evolved over time. Earlier versions can be retained and compared, providing a record of when specific facts, parties,

or claims entered the agent’s knowledge base. In legal practice, where the timing of knowledge can itself be legally significant, this version history has potential evidentiary and procedural value.

Downstream system integration. The structured state lowers the integration barrier for adjacent legal tools and workflows. A downstream system can query the factsheet via a standard database interface to, for example, check for conflicts of interest against other open cases, trigger calendar entries for extracted deadlines, or populate court filing templates with party and claim data. These integrations require structured, typed data; they cannot be built on top of free-form agent responses without a costly and fragile natural language parsing layer.

Summary. The performance metrics reported in Chapter 4 capture one dimension of the architecture’s value: how accurately the agent answers factual questions across a progression of sessions. They do not capture the value of the structured case representation itself—a persistent, inspectable, and reusable artifact that the custom architecture produces as a consequence of its design. In production deployments with real case volumes and real practitioner workflows, this latent value is likely to be at least as significant as the raw performance advantage.

5.3 Uncertainties and Confounding Factors

The evaluation results support the conclusion that the custom agent outperforms the baseline on correctness and relevancy, but several sources of uncertainty affect the magnitude and generalisability of this finding. These are discussed below.

Question difficulty and scope. The evaluation queries were designed to be factual and document-grounded, but the level of specificity varies considerably across sessions. Some questions have unambiguous single-document answers, while others require synthesising information across multiple documents and timelines. Harder queries amplify the architectural advantage of the custom agent — the factsheet and structured retrieval tools are most useful precisely when the answer requires cross-document reasoning — while easier queries may show little to no difference between architectures. The observed aggregate performance gap therefore depends on the composition of the query set: a test set dominated by simple lookups would understate the custom agent’s advantage, while one focused on complex synthesis would overstate it. The current evaluation mix likely sits somewhere in between.

Dataset size and complexity. Only two datasets are used in the evaluation, and their characteristics differ substantially. The results from TOSL-2024, the larger and more complex real case, show a more pronounced advantage for the custom agent than those from THRD-2021. This is consistent with the design hypothesis: the factsheet-based architecture is most beneficial when the document set is large and the case is complex, because those are the conditions under which naive retrieval is most likely to fail. With only two datasets, however, it is difficult to disentangle the effect of document volume from other case-specific characteristics. A larger benchmark spanning more cases and varying document volumes would provide stronger evidence for the scaling behaviour observed here.

Underlying language model capability. Figure 4.4 shows that the performance gap between the custom and baseline agents narrows as the underlying LLM becomes more capable: for the most advanced model tested (Gemini 2.5 Pro), the two configurations are nearly indistinguishable. This suggests two non-exclusive interpretations. First, a sufficiently capable model may compensate for the structural disadvantage of the baseline by inferring missing context or tolerating retrieval noise more robustly. Second, the evaluation queries and datasets may not be sufficiently demanding to expose the limits of a powerful model: the questions can largely be answered from the retrieved chunks even without the accumulated factsheet context. Conversely, the performance gap is most pronounced for the smaller models, for which structured, pre-digested context provides the greatest marginal benefit. This implies that the value of the custom architecture is not fixed but model-dependent: it acts as a capability amplifier that matters most when the underlying model has limited reasoning capacity or context-handling ability.

Stochastic variability. LLM outputs are inherently stochastic. Each evaluation set was run three times to reduce the impact of this variability, and the non-parametric Mann-Whitney U test is robust to the non-normal, left-skewed score distributions observed. Nevertheless, the total number of unique query-session combinations is constrained by the cost and time of running large-scale multi-model evaluations, and the reported p-values should be interpreted with awareness of the limited sample size. The significance of correctness ($p = 8.5 \times 10^{-3}$) and relevancy ($p = 3.4 \times 10^{-4}$) results are robust, but the borderline result for completeness ($p = 8.3 \times 10^{-2}$) should be treated cautiously.

Chapter 6

Conclusion

This thesis investigated whether schema-driven cross-document extraction into a structured case representation — a *FactSheet* — improves the factual accuracy of an agentic RAG system in multi-document legal settings, and whether it does so without incurring proportional increases in query-time cost. The motivation was grounded in a fundamental limitation of chunk-based retrieval: when the answer to a legal query is distributed across multiple documents, no amount of semantic similarity tuning can make an isolated-chunk retriever assemble a coherent synthesis. The proposed approach addresses this by extracting typed entities — parties, events, claims, damages, deadlines — from all case documents at ingestion time and maintaining them as a persistent, relational representation that supplements retrieval for case-level queries.

6.1 Answers to Research Questions

Main RQ. Schema-driven extraction into a structured case representation does improve factual accuracy: the custom agent achieves statistically significant improvements in correctness ($p = 8.5 \times 10^{-3}$) and relevancy ($p = 3.4 \times 10^{-4}$) relative to the RAG-only baseline. Token consumption and inference latency are statistically equivalent between the two architectures, confirming that the quality gain is not purchased at a proportional resource cost. The main research question is answered affirmatively.

RQ1. The evidence for RQ1 is directionally consistent with the hypothesis but insufficient for a firm conclusion. THRD-2021 shows no clear performance gap between configurations. TOSL-2024 shows a somewhat wider advantage for the custom agent in the later sessions, but this rests on few sessions and does not hold uniformly across all metrics. A further theoretical nuance applies: naive RAG is not straightforwardly volume-sensitive, since retrieval always returns a fixed top- K regardless of corpus size. The genuine limitation is query-type specific — synthesis queries whose answers are distributed across multiple chunks — and the evaluation did not explicitly control for this dimension. RQ1 therefore remains open; confirming it would require a dedicated benchmark of synthesis-demanding queries across a larger number of sessions and cases.

RQ2. Session-level trajectories show that the custom agent maintains stable quality with narrow variance across all sessions as document volume grows. The baseline, by contrast, exhibits wider fluctuations throughout case progression. The structured representation scales gracefully with document volume because new information is absorbed incrementally into the factsheet at each session boundary rather than added to an ever-larger unstructured retrieval pool. RQ2 is answered affirmatively.

RQ3. The model-level breakdown reveals that the performance gap between the custom and baseline agents is largest for smaller, less capable models and narrows to near-zero for the most advanced model tested. Structured, pre-digested context reduces the reasoning burden placed on the underlying model, enabling smaller models to approximate the performance of larger ones on the RAG-only baseline. This constitutes a practically significant cost-quality substitution: a cheaper model paired with the custom architecture can match a more expensive model operating on unstructured retrieval alone. RQ3 is

answered affirmatively, with the caveat that the effect is model-dependent and diminishes as base model capability increases.

6.2 Additional Finding: Variance Reduction

Beyond the hypothesised mean improvement, the evaluation reveals a consistent and statistically robust reduction in score variance for the custom agent. Levene’s test for equality of variance is significant across all five metrics tested — correctness, relevancy, completeness, BERTScore F1, and sentence-BERT similarity — including the complementary metrics where mean differences are modest. This finding was not pre-specified as a research question; it emerged from the distributional analysis of the results.

The interpretation is straightforward: pure retrieval is brittle. When relevant chunks happen not to rank highly for a given query, the baseline agent’s performance drops sharply, producing the heavy low-scoring tail visible across all its metric distributions. The custom agent’s structured state provides a stable knowledge base that does not depend on retrieval succeeding for any individual query, dampening these failures and yielding more predictable responses.

In a legal deployment context, this reliability dimension may carry more practical weight than the mean improvement. A practitioner integrating an AI assistant into their workflow needs to be able to predict when the system will and will not perform well. A system with lower variance is easier to trust, easier to supervise, and easier to integrate into structured legal processes than one with comparable average accuracy but unpredictable failure modes.

6.3 Contributions

This thesis makes three contributions:

1. **System design.** A schema-driven initialization and incremental update pipeline for structured legal case state management, implemented as a LangGraph `StateGraph` with fan-out parallel processing, PostgreSQL relational storage, and configurable context compression.
2. **Evaluation framework.** A multi-model evaluation setup comparing a structured-state agent against a RAG-only baseline across two legal datasets, multiple session stages, and three repeated runs per configuration, using a combination of semantic quality metrics (DeepEval correctness, relevancy, completeness), surface similarity metrics (BERTScore), consistency metrics, and resource consumption measures.
3. **Empirical evidence.** Quantitative evidence that structured extraction improves factual accuracy in multi-document legal settings without proportional cost increase, with the effect being strongest on complex cases, stable across session progression, and most pronounced for smaller models.

6.4 Limitations and Future Work

The evaluation rests on two datasets, which limits the generalisability of the findings. A larger benchmark spanning more cases and document volumes would provide stronger evidence for the scaling behaviour observed here, and would better distinguish the effect of document volume from other case-specific characteristics.

The evaluation queries were deliberately restricted to factual, document-grounded questions to enable objective grading. This scope is appropriate for establishing a controlled baseline but does not cover the full range of legal reasoning tasks — in particular, normative and predictive questions where the benefit of structured extraction may differ. Extending the evaluation to cover multi-step legal reasoning, conflicting evidence reconciliation, and temporal argument construction is a natural next step.

The structured representation in this thesis uses a fixed entity schema (parties, events, claims, damages, deadlines). For cases with different legal structures, such as regulatory compliance matters or transactional disputes, the schema may require adaptation. An interesting direction is schema

induction: learning the relevant entity types from the case documents themselves rather than relying on a predefined legal ontology.

The RQ3 result points toward a practically relevant optimization problem: given a target quality level and a budget constraint, what is the optimal combination of model size and extraction investment? The evidence suggests this trade-off exists and is exploitable, but characterising it precisely requires a more systematic sweep over model and architecture configurations than was feasible within the scope of this thesis.

Reliability as a first-class research question. The variance reduction observed across all five metrics is a post-hoc finding and was not designed into the evaluation. A natural follow-up study would treat reliability as a first-class research question: formally hypothesising that structured extraction reduces output variance, designing the evaluation around variance-sensitive query sets (particularly retrieval-failure-prone queries), and testing across a broader range of cases and document volumes. Given the practical significance of predictability in legal settings, this direction warrants dedicated investigation.

Advanced retrieval over structured elements. The current implementation queries the factsheet through simple SQL lookups filtered by date range and significance level. The richness of the extracted representation — typed entities with source references, timestamps, significance scores, and relational links — supports substantially more sophisticated retrieval strategies. Graph traversal across entity relationships, cross-referencing events with the documents that substantiate them, and ranking elements by their relevance to a specific query rather than by static significance are all unexplored. The factsheet as implemented is a data foundation whose retrieval layer has not been fully exploited.

Agentic reasoning and task queuing. The current agent executes a single reasoning pass per query. Adding a task queue that allows the agent to decompose complex queries into ordered sub-tasks — each potentially using a specialised node, toolset, and system prompt — would support multi-step legal reasoning that the current flat architecture cannot perform. This is analogous to chain-of-thought planning at the agent level: rather than producing an answer in one shot, the agent could plan, retrieve, verify, and synthesise across multiple coordinated steps.

Background and asynchronous tooling. The pipeline already runs offline, but the agent itself is purely reactive: it only acts when a query arrives. A natural extension is a set of background tools that run during idle periods — monitoring connected document repositories and inboxes for new material, triggering incremental factsheet updates on arrival, or proactively flagging newly detected deadlines and conflicts to the practitioner. This would move the system from reactive assistant to proactive case monitor.

Graph database and GraphRAG. The relational PostgreSQL schema captures entity types and source links but does not natively represent or traverse relationships between entities. Migrating the structured representation to a graph database would enable direct traversal of entity relationships — for example, following a chain from a claim to the events that substantiate it to the documents that evidence those events. Combined with graph-native retrieval, this would address the cross-document reasoning limitation more completely than the current hybrid approach and bring the system closer to a full domain-specific GraphRAG architecture.

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